

Hub integration progress

Status of the scientific-capability phase: getting the three analysis Hubs running on live data across five NEON sites.

Data Science Institute, UW-Madison · July 17, 2026

CONTAINER

 **Done**

PHASE 0

 **Done**

PHASES 1-5

 **Planned**

DECISIONS

5 pending

Summary

The infrastructure phase is done: a multi-arch container runs standalone CTSM 5.4.043 and completes a NEON tower simulation end-to-end. We are now in the **scientific-capability phase** — getting the three analysis Hubs (Data Analysis, Modeling, Experimentation) running on live/native data across 5 NEON sites. This is a data-rebind and validation effort, not a redesign.

Planning is complete and tracked; the first execution phase (data contract) is done. This report is the shareable status snapshot.

Status at a glance

ITEM	STATUS
Multi-arch container, CTSM 5.4.043	DONE

ITEM	STATUS
KONZ NEON simulation, end-to-end	DONE — 83 monthly files, full 2018-2024 run
Phase objectives / charter	WRITTEN
Implementation plan (6 phases) + tracker	WRITTEN — epic #11
Phase 0 — data contract + reference copies	DONE
Phase 1 — rebind data-access layer	READY — #6
Phases 2-5 — Hubs + validation	PLANNED — #7-#10
Scope decisions with Jingyi	5 PENDING

What we completed this cycle (Phase 0)

- **Data contract** — live CTSM output layout, streams, variables (TSOI, H2OSOI, GPP), dims, and units, grounded in the real KONZ run.
- **5 sample sites confirmed:** KONZ (baseline), ABBY, CPER, TALL, CLBJ.
- **Reference copies received and evaluated** (343 MB, staged locally, gitignored):
 - Right sites, structure, and variables — plug into the existing reader with a path change.
 - **Model output only** — no NEON observations included.
 - **Older model generation** (legacy **h1** / **h0**, 2018-2022) — a shape/plausible-range oracle, not exact ground truth.
- **Naming discrepancy identified:** live output uses **h1a** / **h0a** streams while the current code and reference copies use plain **h1** / **h0**. Phase 1 must handle both — catching this now avoids a silent failure.

Gaps surfaced between the source doc and reality

Mapping `communication-internal` against the code turned up four items needing resolution. Two are features marked "done" in the doc that are actually simpler stand-ins:

FEATURE	DOC SAYS	ACTUAL	TRACKED
ILAMB benchmarking	done	Custom fit metrics (bias, R ² , residuals); no ILAMB	#13
5-step EnKF loop	done	Simple/scalar Kalman filter; no ensemble	#14
Observed data for evaluation	(assumed)	Not in the reference copies; must be sourced	#12
PFT / soil / temperature perturbation	listed	Only precipitation is codified	#3 + MVP scope

These are scope decisions, not missing engineering — except observed-data sourcing (#12), which is a real prerequisite for validating Hub 2.

Decisions pending (Jingyi)

1. **Observed-GPP source** for Hub 2 (#12) — NEON API vs. `evaluation_files` vs. NCAR eval files. *Blocks Phase 3 validation.*
2. **ILAMB** (#13) — accept the custom metrics, or build real ILAMB?
3. **EnKF** (#14) — accept the simple Kalman filter, or build a true EnKF?
4. **Perturbation scope** — precipitation-only for the MVP, with PFT/soil/temp as follow-ons?
5. **Validation tolerance** — how close must live runs match the (older-model) reference copies to pass?

Next steps

- Start **Phase 1** (#6): add a local reader and source-agnostic entry point in `data_access.py` ; fix the `h1` → `h1a` naming for live output. Not blocked by the pending decisions.
- Get decisions 1-3 in front of Jingyi — they set the scope of Hub 2 and whether any net-new work is needed.

Reference documents

- Phase charter — `docs/phase-objectives/`
- Implementation plan — `docs/hub-integration-plan/`
- Data contract — `docs/data-contract.md`
- Execution tracker — GitHub epic #11 (issues #5-#14)